

Transformations

1. CB 2.1 (you do not need to show that the pdf integrates to 1)
2. CB 2.2
3. CB 2.3
4. CB 2.6 (you do not need to show that the pdf integrates to 1)
5. CB 2.7 (a only)
6. CB 2.12
7. CB 2.23
8. Let X be a random variable with pdf

$$f(x) = \begin{cases} \frac{x}{2} & \text{for } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

Let $Y = X(2 - X)$. Find the pdf of Y .

9. Let X be a random variable with pdf

$$f(x) = \begin{cases} \frac{x}{2} & \text{for } 0 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

(this is the same pdf from the previous problem)

Let $Y = 4 - X^3$. Find the pdf of Y .

10. Let X be a random variable with pdf

$$f(x) = \frac{1}{2n+1} \text{ for } x = -n, -n+1, \dots, 0, 1, \dots, n-1, n; \text{ and } 0 \text{ otherwise.}$$

Find the pdf of $Y = |X|$.

Random but Useful

11. CB 3.16
12. Evaluate the following limits (you will need Lemma 2.3.14 for all of these):

$$(a) \lim_{n \rightarrow \infty} \left(\frac{\theta n}{a + \theta n} \right)^{\theta n} \quad (b) \lim_{N \rightarrow \infty} \left(\frac{N-x}{N} \right)^N \quad (c) \lim_{(n-m) \rightarrow \infty} \left(\frac{n-m}{n-m-K+x} \right)^{n-m}$$

* 2.30(d) and 2.38(a) are both asking you to find the moment generating function of the negative binomial distribution (although, 2.30d does not explicitly say it is the negative binomial). Obviously, you only have to derive the mgf once. In order to do so, it might be helpful to use the sum of a negative binomial series:

$$(1-w)^{-r} = \sum_x \binom{x+r-1}{r-1} w^x$$